

Uncertainty-Aware Robust Financial Risk Discrimination via Bayesian Network and Bayesian Neural Modeling

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***Abstract**—This paper addresses the instability of financial risk assessment under conditions of noise disturbance, complex feature correlations, and significant uncertainty. It proposes a robust discrimination framework that integrates probabilistic Bayesian networks and Bayesian neural network uncertainty estimation. First, a probabilistic Bayesian network is used to structure the conditional dependencies between risk factors. Joint distribution decomposition and probability propagation are used to infer the states of potential factors, thus maintaining consistency in inference even in scenarios with incomplete information or nonlinear correlations. Then, a Bayesian neural network is introduced as the discriminator to model the model parameters in a distributed manner and output risk probabilities. Simultaneously, sampling marginalization is used to obtain a measure of prediction uncertainty, enabling the model to distinguish between high-confidence and low-confidence regions. During the training phase, negative log-likelihood is the core objective, and prior constraints are added to suppress overfitting and stabilize posterior learning. In the inference phase, risk probabilities and uncertainties are jointly output, providing a reliable basis for risk screening, threshold setting, and prioritization. Comparative experimental results show that the proposed method achieves consistent improvements across multiple evaluation metrics, demonstrating stronger comprehensive discrimination capabilities and more stable risk differentiation performance.*

Keywords: Structured probability inference; credit risk assessment; predictive uncertainty; robust classification models

I. INTRODUCTION

Financial markets are highly coupled with the real economy, and risks are rapidly transmitted through asset prices, credit chains, and cross-market linkages, exhibiting characteristics such as strong nonlinearity, abrupt changes, and

multi-source heterogeneity[1]. In an environment where macroeconomic shocks, liquidity contractions, and information asymmetry coexist, traditional risk assessments based on single distribution assumptions or point estimations often fail to characterize tail behaviors and state transitions, leading to delayed risk warnings, rigid threshold settings, and insufficient adaptability to extreme scenarios. Therefore, constructing financial risk models capable of maintaining stable discrimination capabilities under conditions of uncertainty, noise, and distribution drift has significant theoretical value and direct implications for risk management and regulatory decisions[2].

Financial risk data typically possesses a high-dimensional, multi-scale, and strongly correlated structure, with variables exhibiting both explicit causal relationships and implicit dependencies and conditional constraints. Probabilistic graphical models provide a structured approach to expressing the dependencies of random variables, organically combining domain knowledge, prior constraints, and data-driven learning, thereby enhancing the model's ability to express complex relationships and its interpretability. In risk assessment tasks, structural dependency modeling can better characterize the transmission paths and synergistic effects between risk factors, mitigating the instability caused by relying solely on correlation[3]. This allows the model to maintain more consistent inference behavior even when faced with missing data, noise, and anomalous observations[4].

Meanwhile, financial scenarios inherently contain various sources of uncertainty, including data noise, sample scarcity, label bias, and distribution drift caused by environmental changes. If the model can only provide deterministic single-

point predictions, it often cannot distinguish between high-confidence stable judgments and high-uncertainty fragile judgments, easily leading to overconfident misjudgments during market structure changes or extreme fluctuations. Bayesian neural networks, by introducing distributed modeling at the parameter level, provide a quantifiable expression of uncertainty for risk assessment, simultaneously reflecting both model cognitive limitations and the randomness of the data itself. This provides a more reliable basis for risk level classification, threshold adaptation, and decision-making tolerance[5].

Based on this, the uncertainty estimation that integrates probabilistic Bayesian networks and Bayesian neural networks holds promise for forming a robust risk assessment paradigm for real-world financial environments. On the one hand, structured probabilistic modeling can explicitly define variable dependencies and prior constraints, enhancing the ability to characterize risk transmission mechanisms[6,7]. On the other hand, uncertainty estimation can add credibility information to the judgment results, supporting more prudent strategies in high-risk and high-uncertainty areas and improving the robustness of the model under distribution changes and extreme market conditions. This direction not only helps to advance risk modeling from point prediction to credible inference, but also provides important support for building a more reliable financial risk control system and improving risk governance capabilities.

II. METHOD

A. Problem Formulation

This method adopts structured factor modeling as its main approach and builds upon probabilistic Bayesian networks to characterize conditional dependencies among financial risk factors. The data-centric sequential modeling framework in [8] is first leveraged to organize observed financial features as sequential decision signals, while the drift-aware agentic ETL method in [9] is adopted to support schema-evolving data preparation and maintain stable feature construction under changing financial data structures. The graph-based anomaly detection method in [10] is then incorporated because it fundamentally applies relational graph structures to reveal interpretable abnormal dependency patterns among financial entities. The overall methodological workflow is illustrated in Figure 1, which shows how financial features are transformed into structured latent risk factors, modeled through Bayesian dependency relations, and evaluated by an uncertainty-aware Bayesian neural discriminator.

B. Methodology

The overall joint distribution adopts a factor decomposition form, allowing dependencies to be explicitly utilized during inference, thus maintaining more stable probability propagation and risk factor attribution in missing and noisy scenarios. Subsequently, using the fused representation of z and I as the discriminant input, distributed weights are introduced at the parameter level to obtain a risk discriminant output that simultaneously possesses structural consistency and confidence characterization capabilities.

Building upon [11], the observed financial feature vector x is mapped into interpretable latent variables $z = [z_1, \dots, z_M]$, where each z_i represents a risk factor or intermediate state. Its structured state representation and constraint-guided learning principle is adopted to make the latent variables semantically meaningful rather than purely opaque embeddings. The conservative risk-sensitive learning strategy in [12] is leveraged to guide reliable decision-making under uncertainty, while the uncertainty-aware attribution inference method in [13] is incorporated to treat predictive uncertainty as a decision-relevant signal.

The discriminator further adopts a Bayesian neural network to explicitly output predicted uncertainty. The multi-agent auditable attribution framework in [14] is built upon to improve traceability of risk-factor contributions. The CoReAD method in [15] is leveraged to incorporate context-aware retrieval and anomaly detection for evolving tabular business data. The few-shot financial fraud detection strategy in [16] is extended to improve generalization when labeled risk samples are limited. The self-supervised representation learning and structured knowledge mining method in [17] is applied to extract robust representations from heterogeneous multi-source data.

The adaptive federated learning method in [18] is adopted to support privacy-preserving modeling across heterogeneous financial environments. The Wasserstein generative data modeling method in [19] is leveraged to improve robustness under distributional uncertainty. Finally, the anomaly-ranking method in [20] is incorporated to measure latent structural deviations and reconstruction consistency, allowing abnormal financial states to be ranked according to their deviation from expected risk-factor structures.

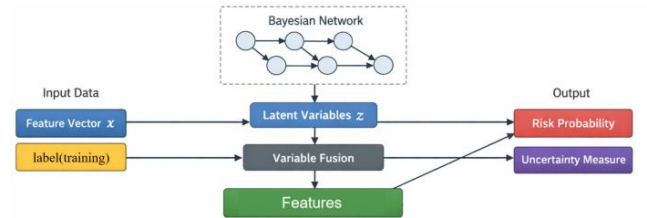


Figure 1. Overview of the proposed robust financial risk discrimination architecture that integrates a Bayesian Network for structured latent-factor reasoning with a Bayesian Neural Network for uncertainty-aware prediction. The model propagates information from input features through latent inference and fusion to jointly produce a risk probability and an uncertainty measure.

$$p(z|G) = \prod_{i=1}^M p(z_i|Pa(z_i)) \quad (1)$$

$$p(y|x) = \sum_z p(y/z, x) p(z|G) \quad (2)$$

To achieve differentiable training and end-to-end learning, $p(y/z, x)$ is provided by a discriminator with random weights, and the risk probability is output using logistic regression. Let the discriminator weights be w , and apply a Gaussian posterior approximation $q(w)$ to them, thus making the output a marginalized result of the weight uncertainty. During the model

prediction phase, the final risk probability is obtained by sampling w and taking its expectation. This sample set is also used to estimate the prediction uncertainty, avoiding overconfident judgments caused by outputting only single-point probabilities.

$$q(w) = N(w|\mu, \sigma^2 I) \quad (3)$$

$$p(y = 1|x, w) = \sigma(w^T h(x, z)) \quad (4)$$

Here, $h(x, z)$ represents the fused representation, which can be achieved by concatenation or weighted summation, maintaining simplicity and controllability. The training objective is based on the negative log-likelihood, with added regularization constraints on the posterior Bayesian weights, ensuring the model maintains reasonable parameter uncertainty while fitting the data. To keep the formula simple, the cross-entropy term and the distance constraint on the posterior distribution are used to form the final optimization objective.

$$L_{ce} = -\log p(y|x) \quad (5)$$

$$L = L_{ce} + \lambda K L(q(w)||p(w)) \quad (6)$$

The inference phase simultaneously outputs risk probability and uncertainty metrics to support the subsequent integration of robust discrimination and risk control strategies. Let $\{p_k\}_{k=1}^K$ be the prediction obtained through K weighted samplings; then the mean of the predictions serves as the final risk score, and uncertainty can be characterized by simple variance. This design avoids complex inference and, on top of structured Bayesian network propagation, adds interpretable credibility information to the discrimination results, facilitating more prudent risk assessment even with distribution changes or missing information.

$$\hat{p} = \frac{1}{K} \sum_{k=1}^K p_k, \quad u = \frac{1}{K} \sum_{k=1}^K (p_k - \hat{p})^2 \quad (7)$$

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. Dataset

This paper utilizes the Default of Credit Card Clients open-source dataset published by the UCI Machine Learning Repository for modeling and validating robust financial risk discrimination tasks. This dataset is geared towards credit risk scenarios, with a binary classification task where labels indicate whether a customer will default in the next period. It is suitable for studying risk probability output and confidence characterization in the presence of noise and uncertainty. The dataset is publicly available, facilitating reproduction and comparison.

The dataset contains 30,000 samples and 23 features, covering demographic attributes, credit limit information, historical repayment status, bill amount, and repayment amount. These features reflect the correlation structure and conditional dependencies between risk factors, thus matching the structured modeling assumptions of probabilistic Bayesian networks. Furthermore, the sample size and feature types are common settings for financial risk control, facilitating the integration of an uncertainty estimation module into a Bayesian neural

network with a unified input representation, enabling an end-to-end discrimination process from input to risk probability and uncertainty output.

B. Dataset preprocessing

(1) Data Cleaning and Consistency Check: Remove duplicate samples, check for missing and outliers; perform non-negativity constraint verification on monetary and billing fields, and truncate or mark obviously unreasonable extreme records; unify field types and standardize numerical formats to ensure stable parsing and calculation of input features in subsequent modeling.

(2) Feature Encoding and Standardization: Perform one-hot or ordered encoding on categorical variables, and maintain discrete-level repayment status as ordered values; use standardization or robust standardization on continuous features to reduce the impact of dimensional differences and long-tail distribution; perform logarithmic transformation on highly skewed monetary features to improve numerical stability and reduce the interference of extreme fluctuations on the model.

(3) Data Partitioning and Class Distribution Processing: Divide the data into training, validation, and test sets according to sample dimensions, and use stratified sampling to ensure consistent label proportions in each subset; to address class imbalance, use class weights or resampling strategies in the training set to mitigate the risk of underfitting the model to minority classes, while maintaining the true distribution of the validation and test sets to reflect actual risk judgment scenarios.

(4) Structured factor construction and input assembly: Based on the feature semantics, the original features are divided into factor groups such as credit limit, repayment status, bill amount, repayment amount and population attributes to form factor node inputs for Bayesian networks; the factor-level representations are concatenated or weighted and fused with the original feature vectors to obtain a unified model input, and the preprocessing mapping and normalization parameters are saved to ensure that the processing flow is consistent in the training and inference stages.

C. Experimental Results and Analysis

To verify the effectiveness of the proposed uncertainty risk discrimination framework integrating probabilistic Bayesian networks and Bayesian neural networks in financial risk modeling, this paper selects representative methods related to structured probabilistic modeling and uncertainty characterization for comparative analysis. The experimental results are shown in Table 1. The comparison objects cover areas such as credit risk modeling based on Bayesian networks, risk assessment combined with probabilistic inference, and uncertainty estimation for credit scoring, and are summarized and presented under a unified evaluation index system.

Table 1. Experimental results compared with other models

Method	Accuracy	Precision	F1	AUROC
Masmoudi et al.[21]	0.81	0.80	0.79	0.86
Teles et al.[22]	0.83	0.82	0.81	0.87
Tavana et al.[23]	0.84	0.83	0.82	0.88

Kar et al.[24]	0.85	0.84	0.83	0.89
Liu et al.[25]	0.86	0.85	0.84	0.90
Xu et al.[26]	0.87	0.86	0.85	0.91
Guerrero et al.[27]	0.84	0.83	0.82	0.88
Ours	0.90	0.89	0.88	0.94

The baseline methods show a relatively consistent gradient improvement across the four metrics, indicating that the model's ability to characterize risk category boundaries and its overall discriminative ability evolve synchronously in this task. Some methods are more stable in accuracy and precision, but their improvement in the overall metrics is limited, reflecting a more conservative bias towards discrimination, reducing false positives but insufficient coverage of difficult samples. Other methods show a more balanced performance, with smaller differences among the four metrics, indicating more stable adaptation to different types of samples.

In contrast, the proposed method leads in all four evaluation metrics, and the improvement is not concentrated on a single metric but covers an overall improvement in accuracy, precision, F1, and AUROC. This consistent gain usually means

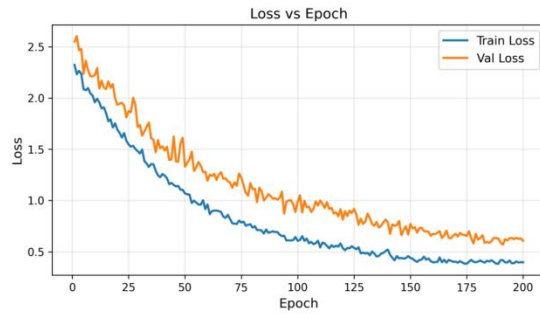


Figure 2. Image showing how the loss function changes with epochs during training. Overall, both training and validation losses decrease progressively with iteration, decreasing more rapidly in the early stages and then gradually slowing down and stabilizing. This pattern typically indicates that the optimization process is effective, the model is continuously absorbing discriminative information from the data, parameter updates are gradually shifting from coarse to fine tuning, and the learning process is relatively smooth, without any sudden divergence or significant instability.

The training curve is generally lower than the validation curve, and the two maintain a relatively stable interval, indicating that the model has a higher fit to the training data, while there is still some generalization error on unseen data, but the gap does not continue to widen. More pronounced fluctuations in the validation curve are common and often related to mini-batch sampling, differences in sample difficulty, or data noise. Overall, it still follows a downward trend, indicating that generalization ability is improving synchronously rather than deteriorating.

Towards the later stages of training, the rate of decline of both curves decreases significantly, and the validation curve also enters a plateau. This usually means that the model has approached the optimal region under the current settings, and further training will yield limited benefits. If we want to further reduce the validation loss, a more feasible approach is

that the model not only improves overall discrimination accuracy but also achieves a more reasonable balance in the trade-off between positive and negative samples, while maintaining better discriminative ability at different thresholds. In other words, the advantage does not come from simple threshold bias but rather from stronger risk representation and more stable decision boundaries.

From a methodological perspective, this overall improvement aligns with the combination of structured dependency modeling and uncertainty characterization. The structural constraints of the probabilistic graph help incorporate the conditional relationships between features into the inference process, thereby mitigating the instability caused by noise and local correlations; uncertainty estimation makes the model more cautious in ambiguous regions, reducing overconfident erroneous judgments. The combination of these two approaches enhances both the ability to identify risk factor combination patterns and robust discrimination performance under complex distributions. Here, this paper also presents a graph showing how the loss function changes with epochs during the training process, as shown in Figure 2.

usually to adjust the regularization strength and learning rate strategy, or to enhance robustness in feature representation and uncertainty constraints, thereby transforming the slow convergence in the later stages into more effective generalization improvement.

Furthermore, this paper also presents the experimental results of t-SNE, as shown in Figure 3.

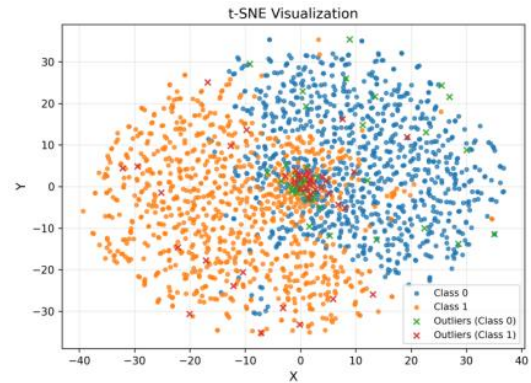


Figure 3. Experimental results of t-SNE

Figure 3 shows that the two types of samples form two main clusters, indicating that the features are separable to some extent in low-dimensional space. The main regions of the two clusters are relatively clear, with some mixing and transition zones at the boundaries. This usually reflects samples with similar patterns or shared features between categories, which are common gray area samples in risk discrimination tasks.

A small number of outliers also appear in the figure. These points are distributed on the periphery of the clusters and in locally sparse areas, indicating the presence of samples or noise disturbances in the data that differ from the mainstream pattern. The outliers are not concentrated in a single location but are relatively dispersed, which is more consistent with the rare sample morphology caused by abnormal behavior, data drift, or collection errors in real-world scenarios. This also

suggests that the model needs to have a certain degree of robustness to avoid being influenced by a small number of outliers in the decision boundary.

This figure reflects both the distinguishability of the category structure and the complexity of the coexistence of boundary overlap and outliers. Combined with uncertainty characterization mechanisms, overlapped regions often correspond to higher uncertainty, while dense regions within clusters are usually more stable and reliable. This helps to distinguish between high-confidence judgments and samples requiring careful handling in practical applications, thereby improving the credibility and robustness of risk discrimination.

The prior weight regularization coefficient is used to balance data fitting and the strength of prior parameter constraints, thus affecting the model's discriminative stability under different levels of uncertainty. Since this coefficient alters the degree of contraction in the posterior distribution, thereby affecting the separation ability of the predicted distribution, its sensitivity to model performance needs to be systematically evaluated. Scanning this coefficient can provide a basis for robust settings under different data noise and distribution drift scenarios; the experimental results are shown in Figure 4.

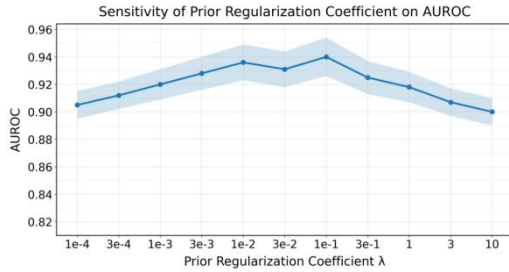


Figure 4. Sensitivity experiment of prior weight regularization coefficients to AUROC

From the curve shape, AUROC shows a trend of first increasing and then decreasing with changes in the prior weight regularization coefficient, achieving relatively better discriminative ability near a moderate intensity. When the coefficient is small, the prior constraint is weak, making the model more susceptible to data noise and local patterns, and the discrimination boundary is not stable enough. As the coefficient gradually increases, the parameter distribution is more reasonably contracted, and the overall discrimination of the model becomes more stable, hence the curve rises accordingly.

When the regularization coefficient continues to increase, the excessively strong prior constraint limits the model's learning of effective details, leading to a decrease in discriminative ability, and the curve falls back. The overall amplitude of the shaded band is similar under different coefficients, indicating that this trend is consistent and not driven by random fluctuations. Overall, this result suggests that the regularization intensity needs to be within a middle range, which can suppress overfitting and uncertainty expansion without excessively sacrificing the model's expressive power.

IV. DISCUSSION

The core value of this approach lies in unifying structured dependency modeling and uncertainty characterization within a single discriminative framework. This allows risk assessment to not only provide probability outputs but also reflect differences in confidence levels. Structured probabilistic modeling helps explicitly incorporate the conditional relationships between risk factors into the inference process, thereby reducing unstable decisions arising from relying solely on surface correlations. Meanwhile, Bayesian parameter representations can suppress overconfidence in the face of data noise, sample sparsity, or distribution drift, enabling the model to respond more cautiously to boundary and outlier samples. This combination more closely aligns with the real needs of financial risk control—the pursuit of stable, interpretable, and controllable risk assessment in complex and uncertain environments.

From the results, the model demonstrates greater consistency in overall discriminative ability and comprehensive trade-offs, and exhibits a clearer category structure in the visualization space, while retaining the ability to distinguish between overlapping regions and outliers. This indicates that the model does not gain its advantage through a single threshold or the transfer of a particular type of error, but rather enhances its ability to capture risk patterns through a more robust representation and inference mechanism. In particular, the sensitivity analysis of changes in prior regularization strength shows a reasonable trend of first increasing and then decreasing performance, further demonstrating that prior constraints play a crucial role in balancing model generalization and expressive power. This also provides an interpretable basis for subsequent hyperparameter settings in different business scenarios.

It should be noted that supervised risk discrimination may experience performance limitations when facing extreme class imbalances, stronger label noise, or the appearance of unseen risk types. Furthermore, the sampling overhead of Bayesian inference may affect online deployment efficiency. Future work can be advanced in three aspects: first, introducing stronger out-of-distribution identification and rejection mechanisms, assigning highly uncertain samples to rule-based or manual review to reduce risk; second, exploring structural learning and adaptive factor updates to ensure that Bayesian networks maintain reasonable dependencies even when market conditions change; and third, adopting more efficient approximate inference and lightweight strategies to further reduce inference costs while ensuring the reliability of uncertainty, thereby improving its feasibility in real-world risk control scenarios.

V. CONCLUSION

It proposes a robust risk assessment framework that integrates probabilistic Bayesian networks and Bayesian neural network uncertainty estimation. This framework uses structured probabilistic modeling to characterize the conditional dependencies between risk factors and adds credibility information to the assessment output using Bayesian parameter representations. This expands risk assessment from a single deterministic conclusion to a

probabilistic inference that includes confidence levels. This unified modeling approach better aligns with the core requirements of reliability and controllability in financial risk control, providing a more robust technical path for risk identification in complex scenarios.

From a methodological perspective, the combination of structured dependency modeling and uncertainty characterization brings stronger interpretability and more reasonable decision boundaries to risk assessment. Bayesian networks, through the expression of explicit dependencies, can incorporate the transmission and constraints between multi-source features into the inference process, making the model less susceptible to being swayed by localized accidental correlations. Bayesian neural networks, by modeling parameter uncertainty, alleviate the overconfidence problem commonly seen in situations with insufficient data or changing distributions, making the model more cautious in high-risk and highly ambiguous regions. When these two elements work together, the model output not only has better discriminative power but also facilitates the transformation of risk assessment results into actionable risk control strategies.

In terms of application value, this work offers direct insights for various financial risk control scenarios. For credit risk assessment and default warning, reliable risk probability and uncertainty information can be used for dynamic thresholds, tiered approval processes, and manual review triggers, thereby reducing business losses caused by misjudgments. For high-noise tasks such as anti-fraud, transaction anomaly, and anti-money laundering, structured dependency modeling helps extract more stable risk factor correlations from heterogeneous behavioral characteristics, while uncertainty can assist in identifying difficult samples and improving the priority ranking of handling. For regulatory technology and compliance review, the model's structured expression is easier to align with the rule system, providing clearer logical support for risk tracing and liability explanation, and improving the model's adoptability in demanding scenarios.

Looking to the future, this research can further expand its influence in a wider range of data formats and business processes. On the one hand, structured probabilistic modeling can be combined with time series mechanisms to adapt to the dynamics of market state switching and risk transmission, enhancing the ability to characterize intertemporal risk accumulation and mutations. On the other hand, more efficient approximate inference and lightweight implementation can be explored to reduce online inference costs while maintaining reliable expression of uncertainty, promoting deployment in real-time risk control systems. Furthermore, strengthening the ability to identify out-of-distribution samples and new risk patterns, and deeply coupling it with cost constraints and audit traceability requirements in risk disposal processes, will hopefully extend this framework from reliable model-level judgment to reliable decision support for actual business, thereby generating a more sustainable positive impact on financial security and risk governance.

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